

① Variables

A Variable is a named container that stores a value in memory, which can be used and changed by the program and it has a fixed type.

How to use:

dataType variableName = value;

Ex: int age = 21;

When to use:

Whenever you need to store data (input, results, values) to use or change later.

Where to use:

Inside methods, loops, conditions, classes - anywhere in a program.

Why declare a type:

- Java is statically typed
- Catches errors early, allocates memory efficiently, keeps large code bases safe.

Real-life Example: like a labeled-jar, the label (name) stays the same, contents (value) can change.

Naming Rules:

- Case Sensitive
- must start with (letter, \$, _) not a digit.
- convection: camel case (Ex: studentName).

Type of Variables:

<u>Type</u>	<u>Declared</u>	<u>Scope</u>
Local	Inside a method	Only within that method
Instance	Inside class, outside methods.	Belongs to each object
Static	Inside class, with static keyword	Shared across all objects.